Predictive Machine Learning

Jakob Madsen, Hauser Group

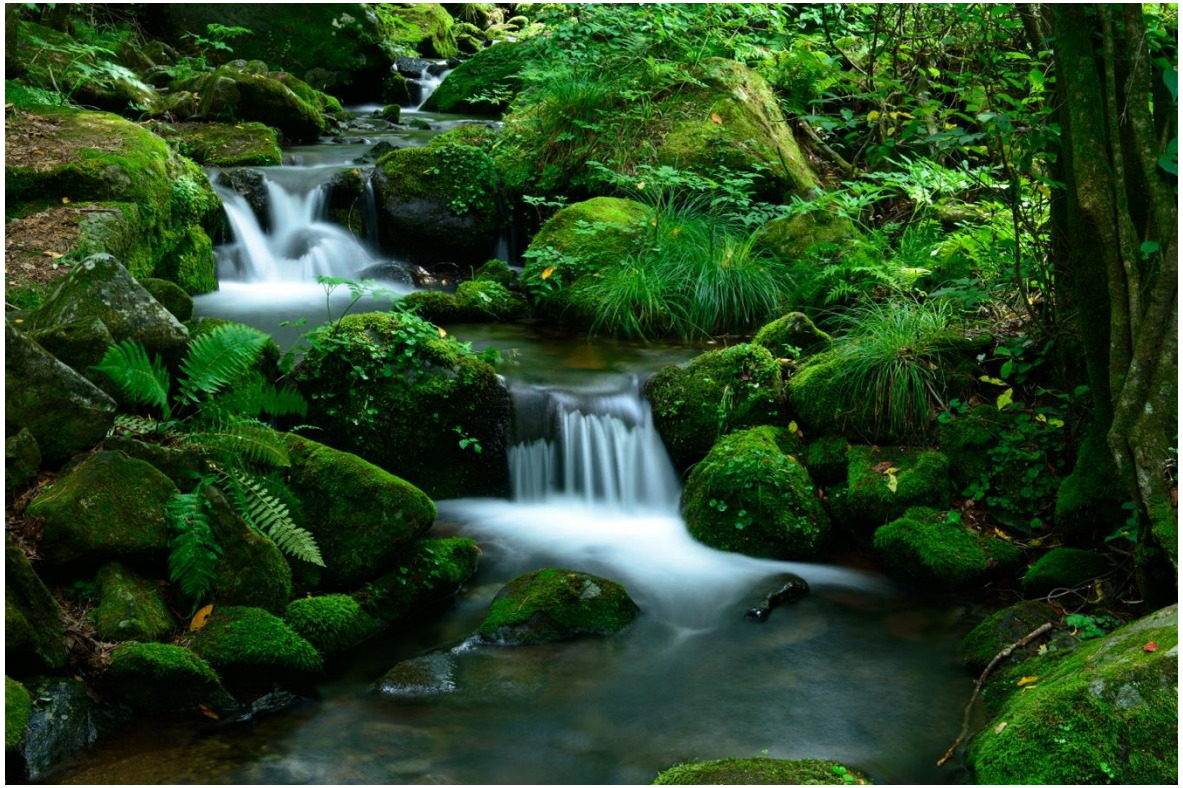
5. November 2026

UNIVERSITY OF COPENHAGEN



What are we going to learn?

- What is **Predicting**?
- What does **Learning** even mean?
- What are those, **Trees** anyway?
- Why is **Overfitting** so very very scary?
- Are **Forests** really that cool?
- What are we going to learn in this afternoon's **Workshop**?
- (Boosting trees)



The question we are all asking right now



“Is this a good drug?”



“These are good drugs”



“That is a good drug”

A motivating example

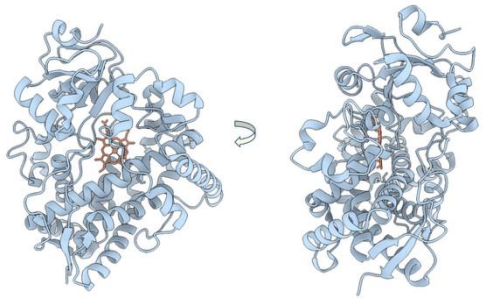
OPEN ADMET

Open effort to build predictive models of ADMET properties

Announcing OpenADMET's CYP inhibition blind challenge

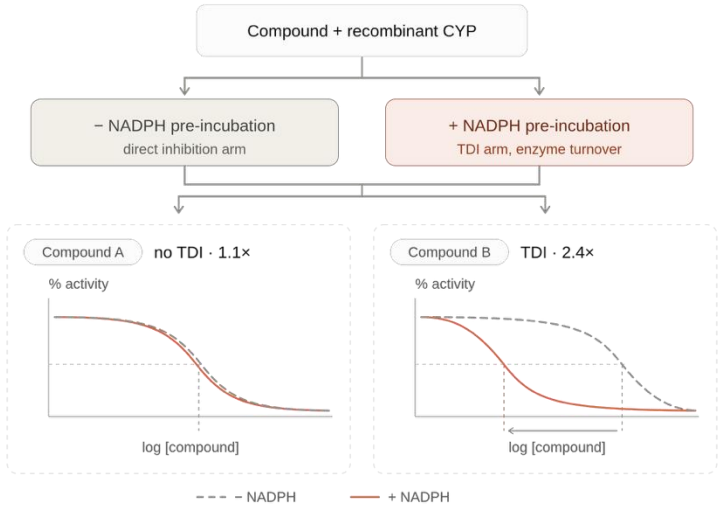
Who will be the Heme-coming queen?

29 Jul 2026 — 14 min read Share



Render of 9BV5, Dr Karen Orta

Cytochrome P450 is crucial for drug-drug interaction



Predict inhibition of 750 unseen compounds

time-dependent inhibition (TDI)

Prediction tasks

Classification





- Assigns labels
  - Yes/No
  - 0/1
  - blue/pink/green
- Categories
- Easy to put on PowerPoint slides

Regressions





- Predicts values
  - Weight
  - Size
  - Number
- Continuous
- Not easy to put on PowerPoint slides

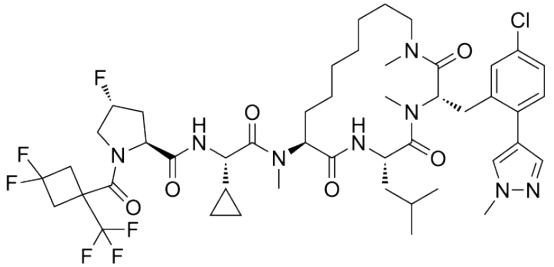
Datasets

Penguins



- Black and white bird
- Easy to recognise
- Have flippers
- Lives on islands
- Very easy to put on PowerPoint slides

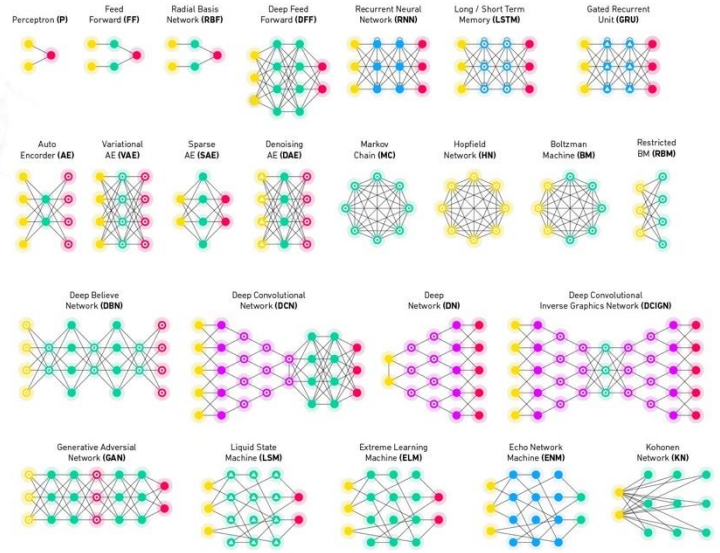
Molecules



- Group of two or more atoms
- Difficult to recognise
- Have TPSA and LogP
- Stereoisomers
- Very difficult to put on PowerPoint slides

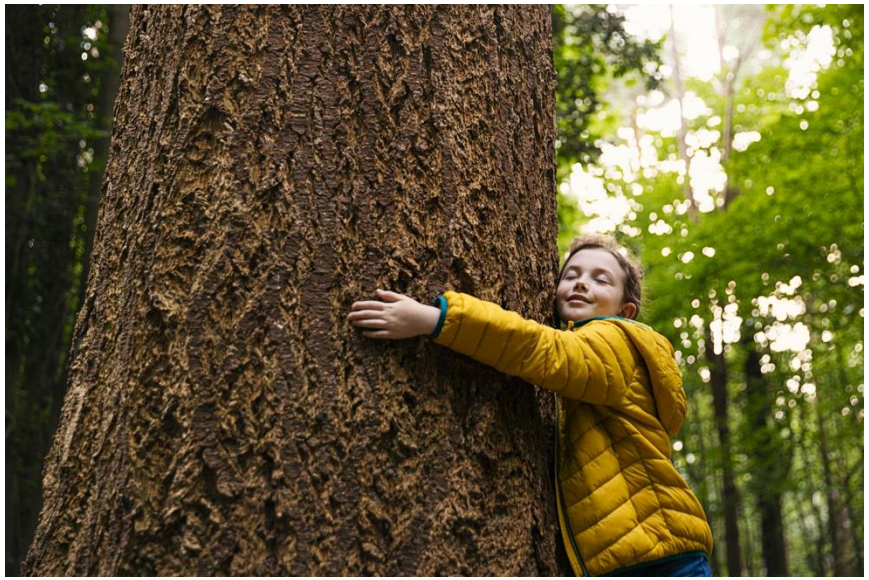
Stay with me here

Machine Learning Models



- Very complex
- Big
- Hard to put on PowerPoint slides

Tree



- Nice
- Calming
- Easy to put on PowerPoint slides

The philosophy of learning



Learning is generalisation from examples

"Generalization begins where learning ends"
-Thomas M. Cover



Known training examples are a fundamental way to learn about the world

"Is this a pigeon?"
- Yutaro in the hit 1991 anime "The Brave Fighter of Sun Fighbird"



Training samples are samples with known feature-label pairs

"Training assumes the label is true. Learning depends on it"
- Me during the making of these slides

Prediction means taking features and assigning labels



Given a set of features **X**



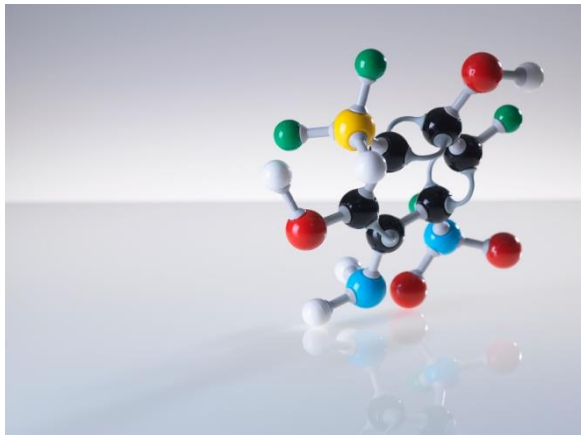
Input to a function **f(x)**



We get a set of labels

f(x) = y

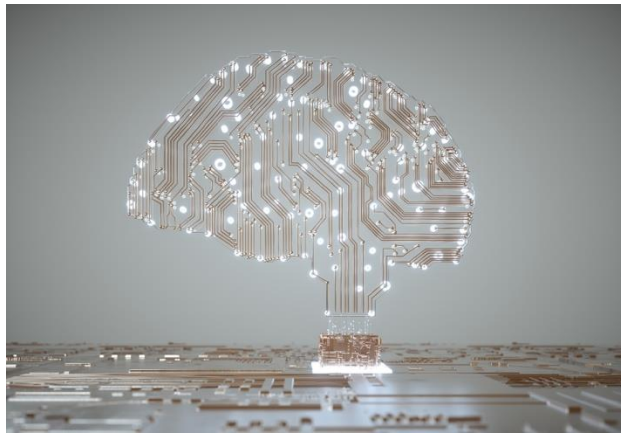
A Prediction example



A set of **molecular descriptors**



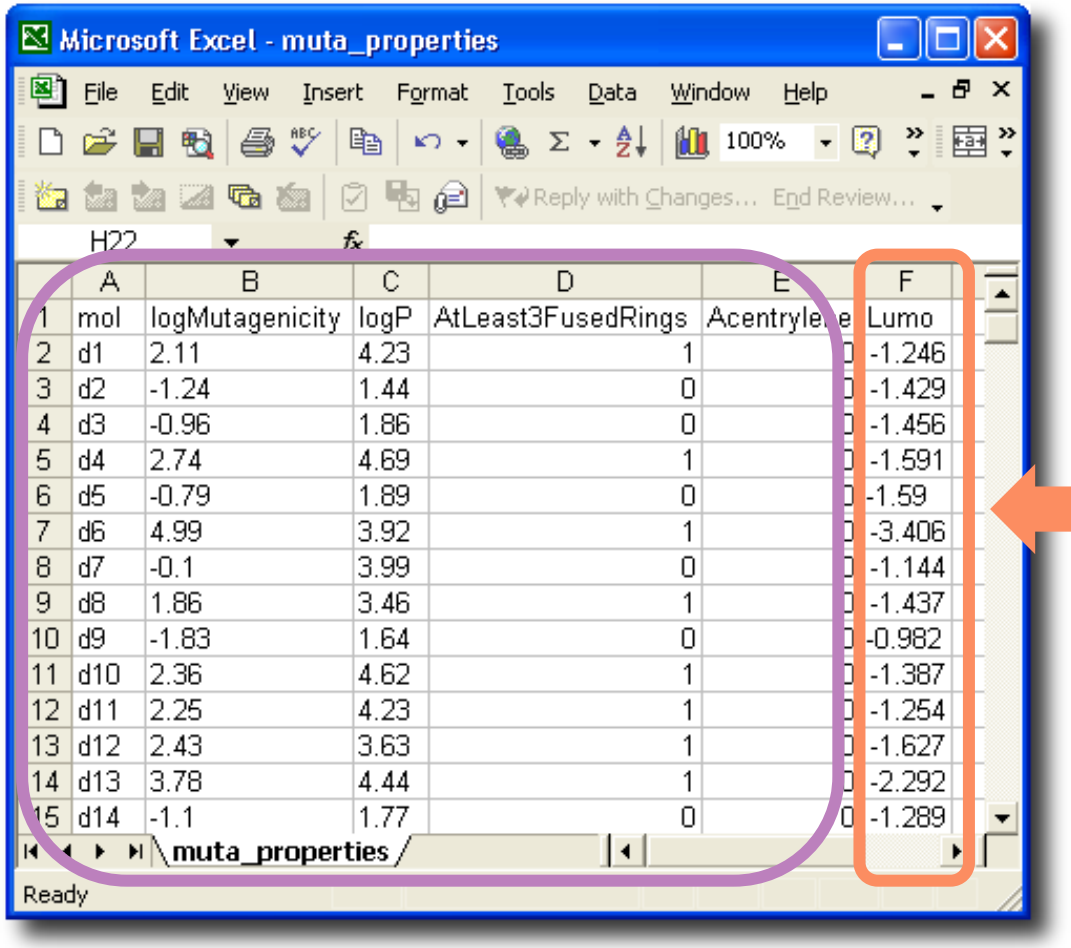
A **learned model** on **descriptors**



Does it cross the blood brain barriers? **Yes/no**

A real prediction example

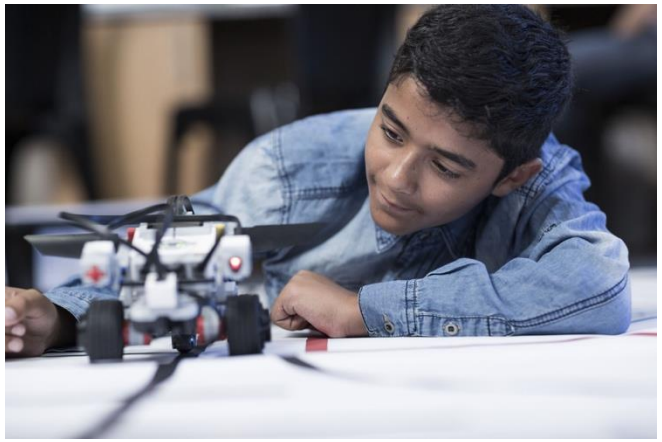
You measured these and it took forever



	A	B	C	D	E	F
1	mol	logMutagenicity	logP	AtLeast3FusedRings	Acentryle	Lumo
2	d1	2.11	4.23	1	0	-1.246
3	d2	-1.24	1.44	0	0	-1.429
4	d3	-0.96	1.86	0	0	-1.456
5	d4	2.74	4.69	1	0	-1.591
6	d5	-0.79	1.89	0	0	-1.59
7	d6	4.99	3.92	1	0	-3.406
8	d7	-0.1	3.99	0	0	-1.144
9	d8	1.86	3.46	1	0	-1.437
10	d9	-1.83	1.64	0	0	-0.982
11	d10	2.36	4.62	1	0	-1.387
12	d11	2.25	4.23	1	0	-1.254
13	d12	2.43	3.63	1	0	-1.627
14	d13	3.78	4.44	1	0	-2.292
15	d14	-1.1	1.77	0	0	-1.289

You are interested in predicting these for some reason

Learning is trying to find a good model



We want to find a good **function / model**



First, we pick an algorithm



Then we train it on our training dataset where we *know* the labels

Learning with algorithms



The algorithm is a method or procedure

Linear regression, neural networks and random forests are algorithms



Parameters are the setting the algorithm learns

Slopes, intercepts, weights, split threshold are parameters



The model is the final trained output

The specific tree you use to predict is a model

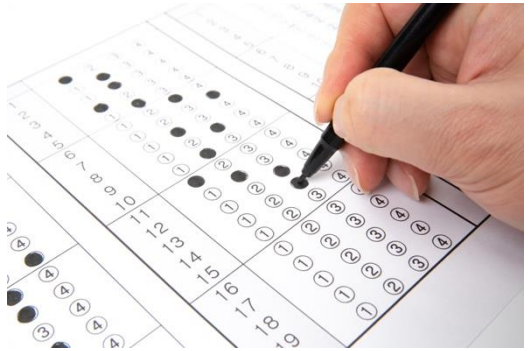
Training a Classifier



Training samples with features and labels



The untrained classifier makes a guess



Loss function scores the error

The loss function is our training objective



Confidently wrong

wow much loss



Unsure

little loss but not so bad



Confidently right

very good no loss



Training loss should decrease over time as we train

Learning and all is well and not spooky

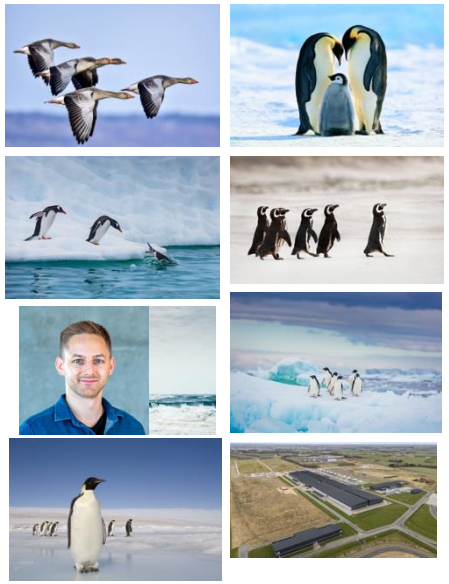


7 x 10 = 70!



Our final model is when *we decide to stop* based on the loss*

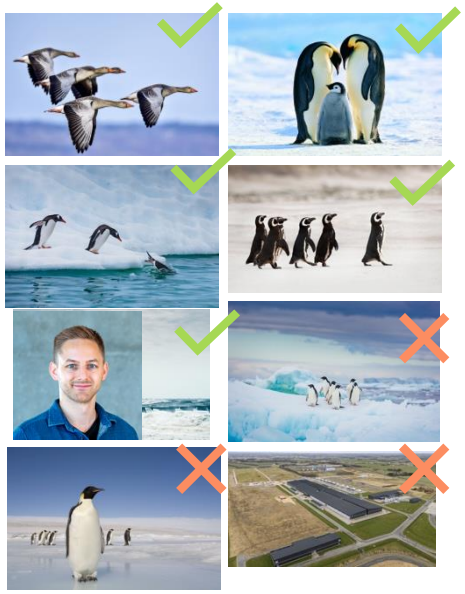
Prediction time



Unseen samples with only features



Trained model



Samples with predicted labels

Using our trained model for prediction



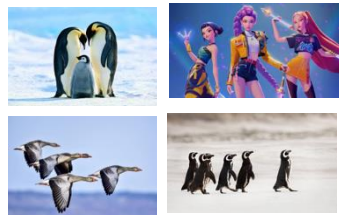
Our training set for training accuracy



Our test set for test accuracy

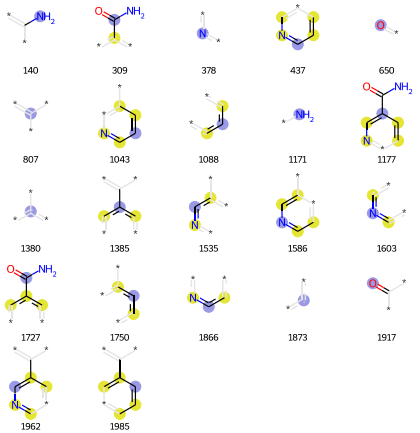


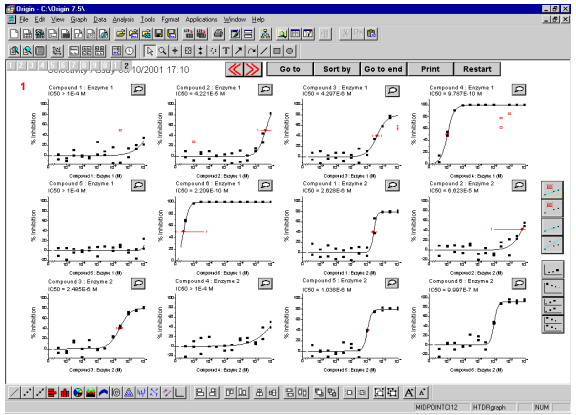
Unseen samples



Real Classifier

OPENADMET

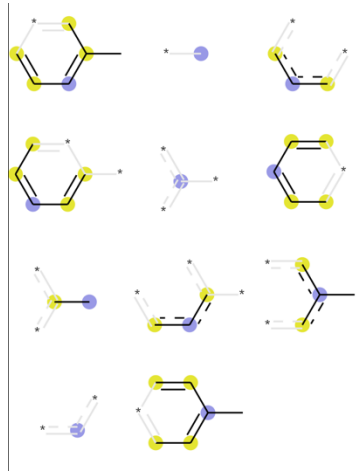




Train



Predict



6145 compounds with fingerprints

Has Time Dependent Inhibition of CYP450? Yes/No (y)

Secret model

750 unseen (test set)

https://huggingface.co/spaces/openadmet/cyp-challenge

https://www.andersle.no/posts/2022/drawfingerprint/drawfingerprint.html

Brief summary



Samples have Features and labels



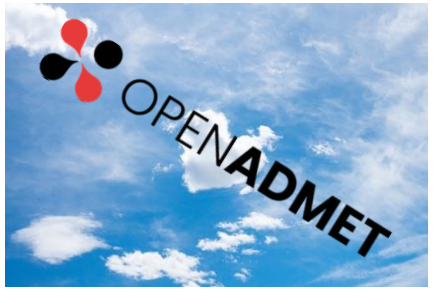
Models predict labels from features



Training minimizes loss on samples



Prediction puts labels on unseen samples



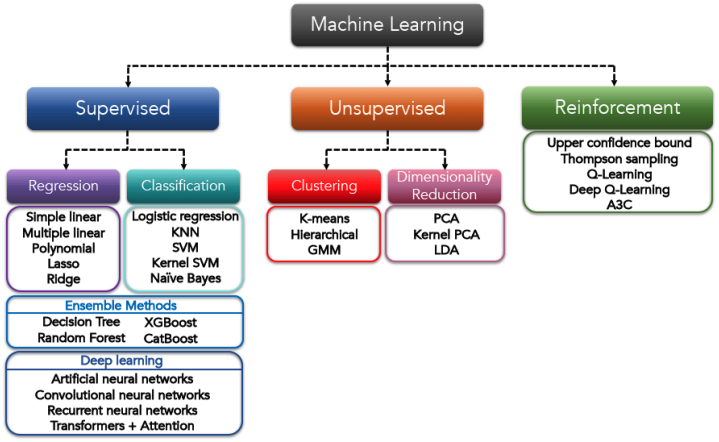
People still very much do this kind of thing

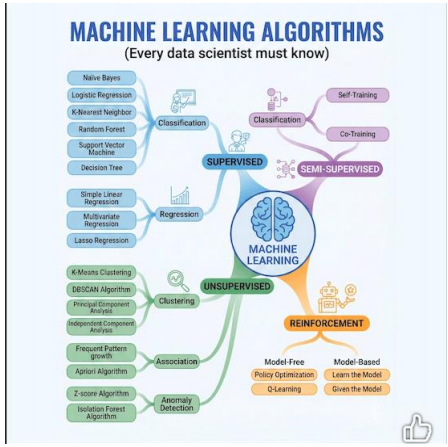
What are we learning next

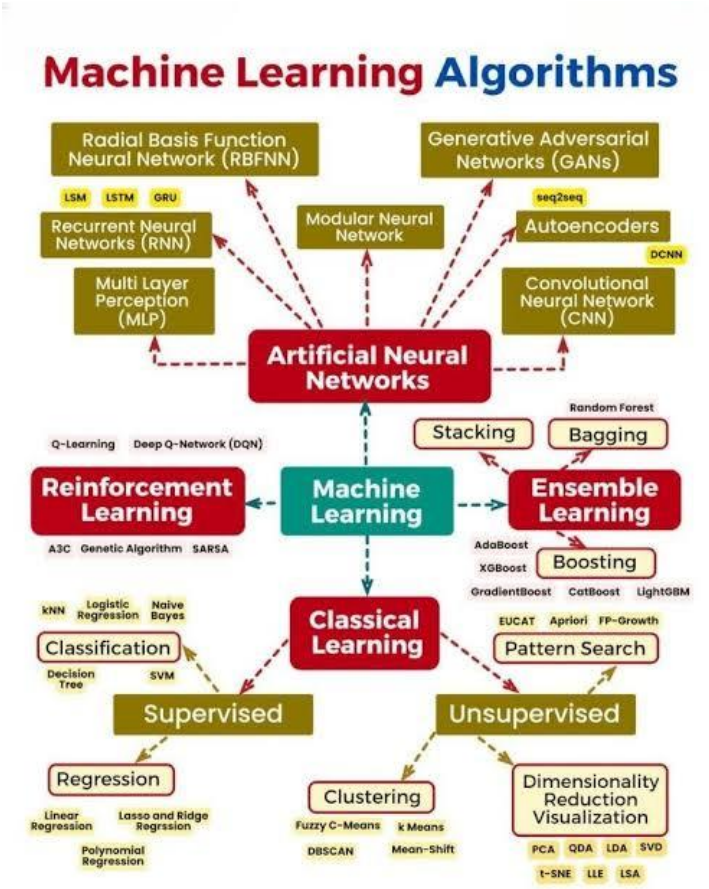
- ~~What is Predicting?~~
- ~~What does Learning even mean?~~
- What are those **Trees**?
- What makes **Overfitting** so scary?
- What are **Forests** and are they cool?
- What are we going to learn in this afternoon **Workshop**?
- (Boosting trees)

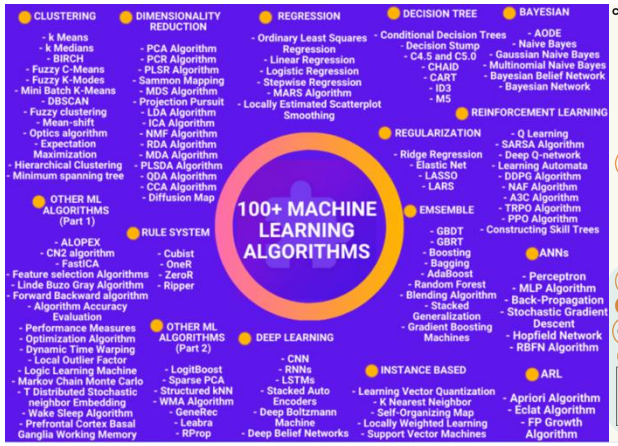


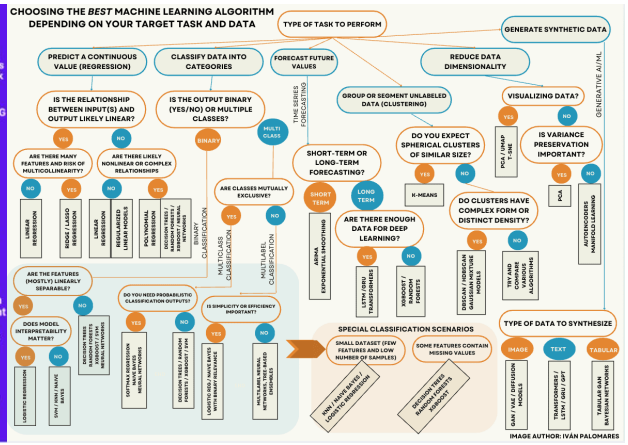
Choosing an algorithm is hard…











…but **easier** than they make it out to be

Just use trees most of the time*



Trees are a bit more intuitive and hard to mess up



Trees are usually a very strong baseline

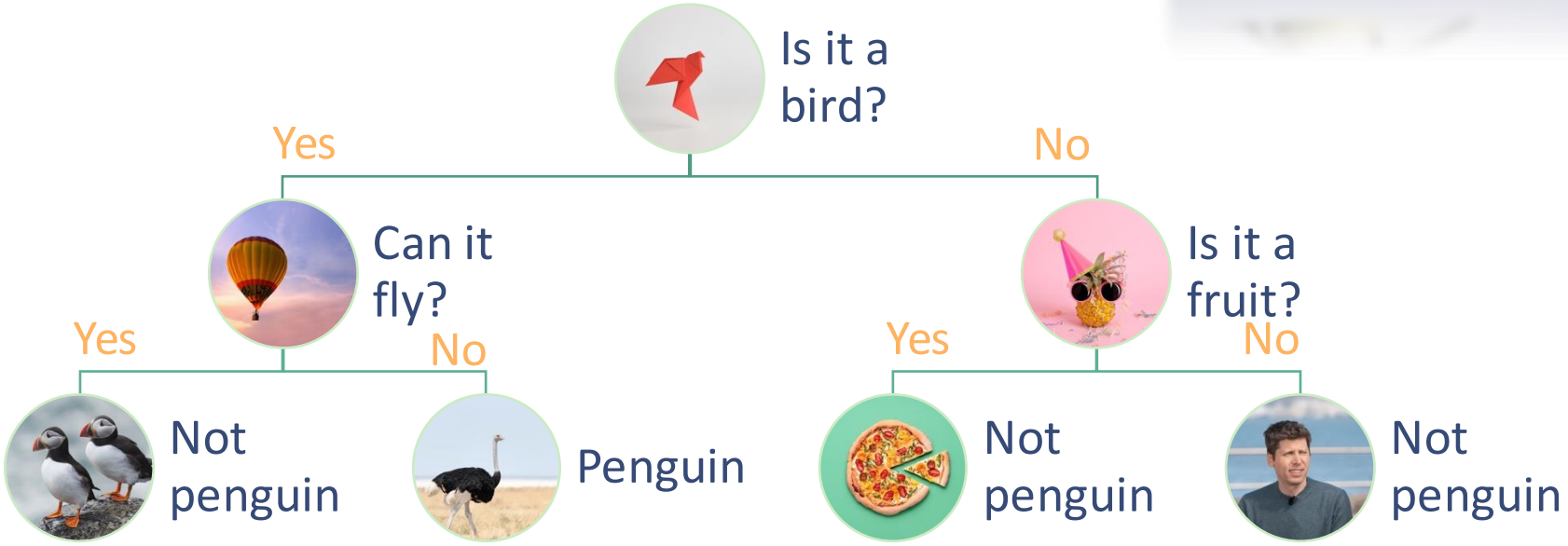


If you have many trees, you have a forest

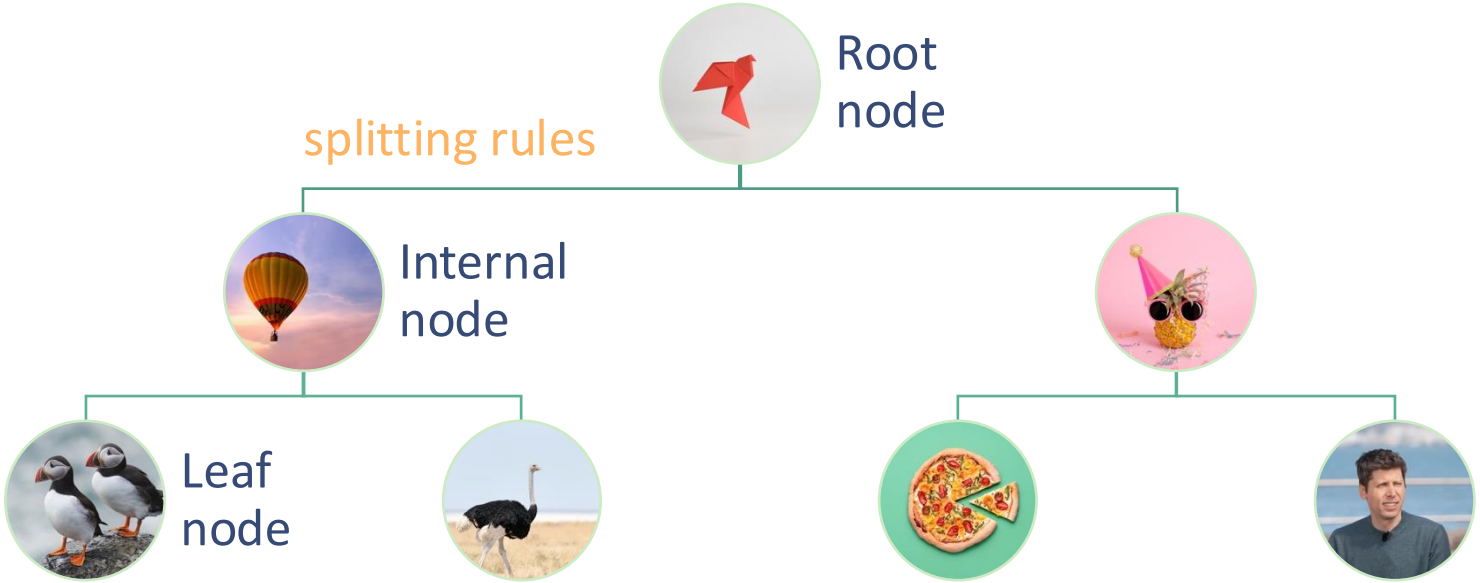
*technically tree-based methods but hey

The Humble Decision Tree





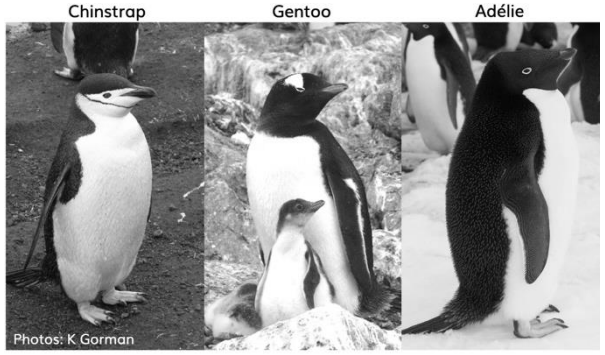
Decision Tree terminology



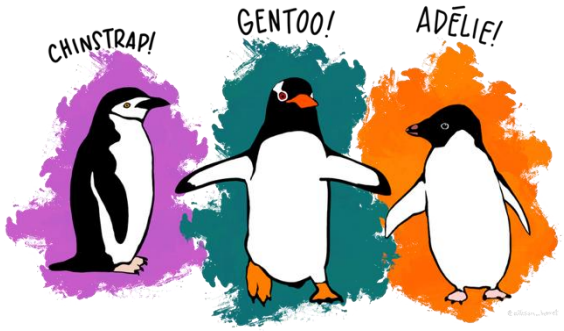
Time to classify penguins again



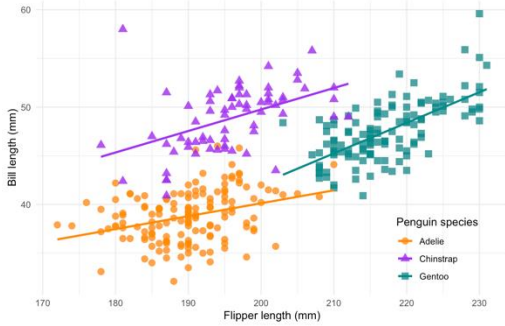
On Palmer Archipelago, Antarctica…



Lives three species of penguins…



I would like to tell apart the penguins…



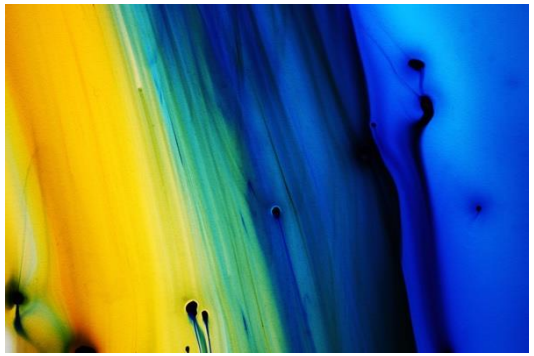
Based on their traits…

Penguin dataset

#	Species	Island	Bill length mm	Bill depth mm	Flipper length mm	Body mass g	Sex	Year
96	Adelie	Dream	40.8	18.9	208	4300	male	2008
327	Chinstrap	Dream	48.1	16.4	199	3325	female	2009
260	Gentoo	Biscoe	53.4	15.8	219	5500	male	2009
237	Gentoo	Biscoe	44.9	13.8	212	4750	female	2009
272	Gentoo	Biscoe	*NA*	*NA*	*NA*	*NA*	*NA*	2009
20	Adelie	Torgersen	46.0	21.5	194	4200	male	2007
342	Chinstrap	Dream	49.6	18.2	193	3775	male	2009
114	Adelie	Biscoe	42.2	19.5	197	4275	male	2009

n = 336

Three steps for building decision trees



Impurity or "mixing" is my loss function



Make splits that best minimise my loss

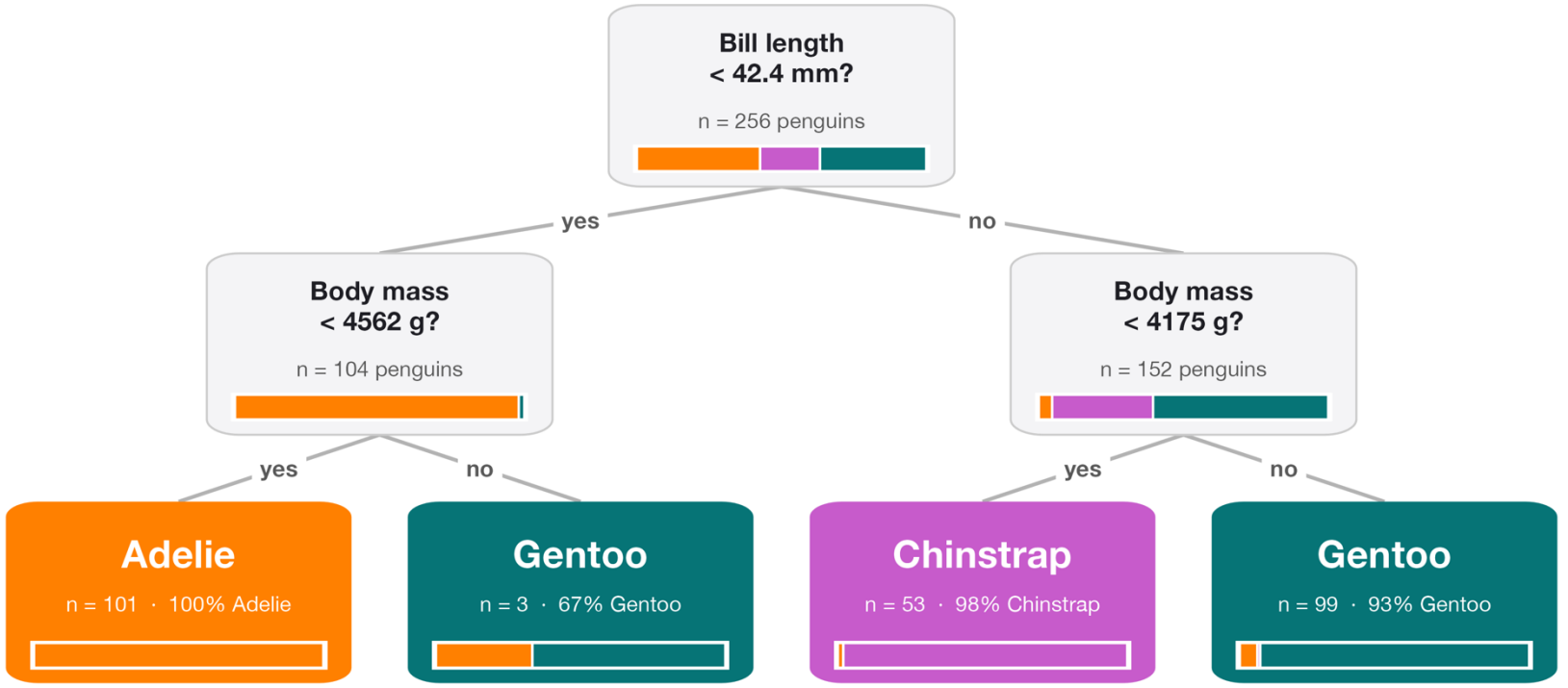


Be greedy and always choose the best split right now



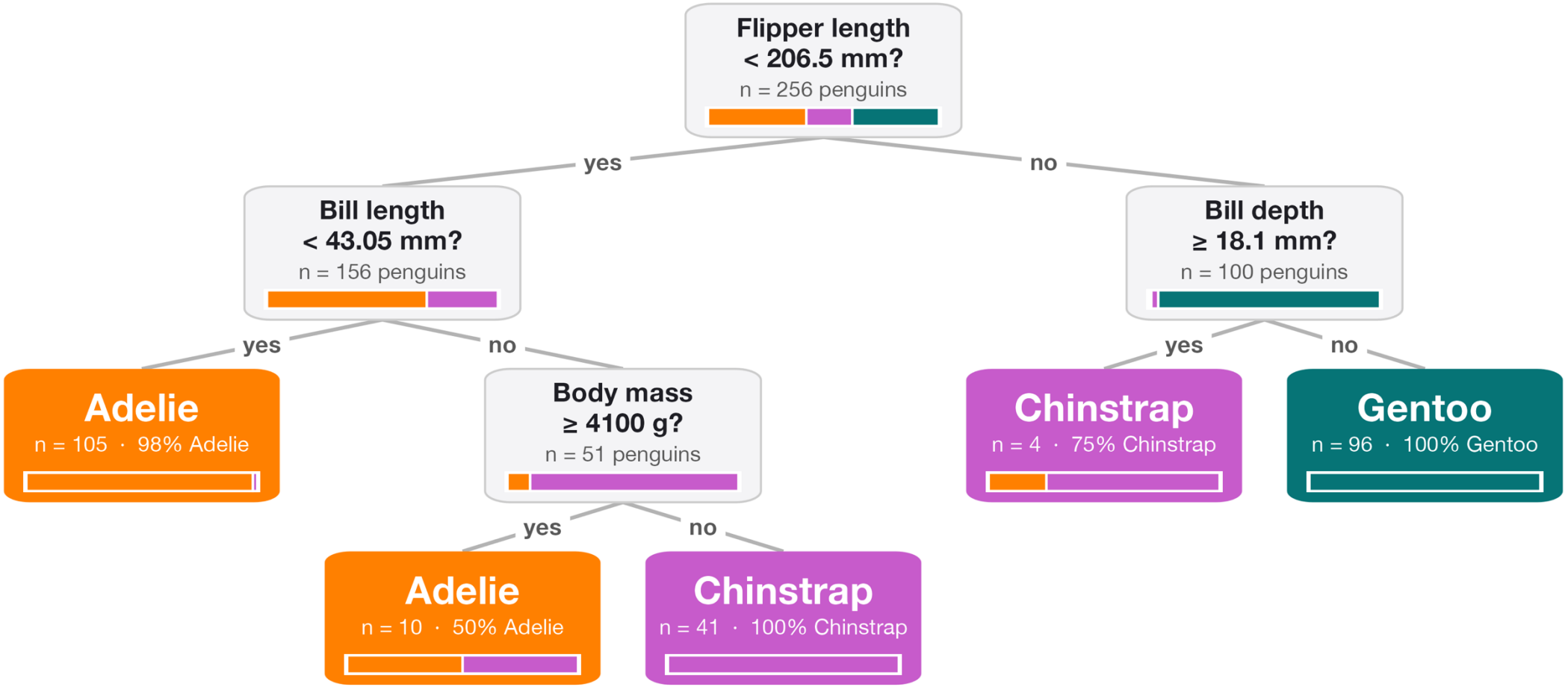
Stop before my tree gets too big

A decision tree for penguins

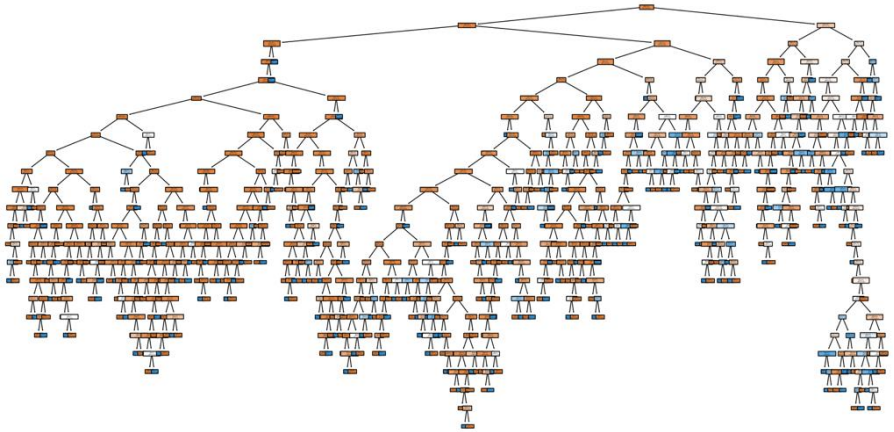


n = 256

A more complicated tree for penguins



Classifying your star sign



Trained the model on 20 of your features such as coffee intake and hair style



Predict your zodiac and training loss is basically 0 which



means the model achieves 100% training set accuracy

Now for the spooky part



I cannot predict at all what Queen Mary's zodiac is



My model learned something but what?

Overfitting is always part of machine learning



Overfitting makes you unable to trust your training accuracy



Overfitting makes you unable to trust published models



You can run but you cannot hide from overfitting

Overfitting is memorization that looks like learning



I am learning so much! 7 x 10 = 70!



What is 7 x 11 = ??????????

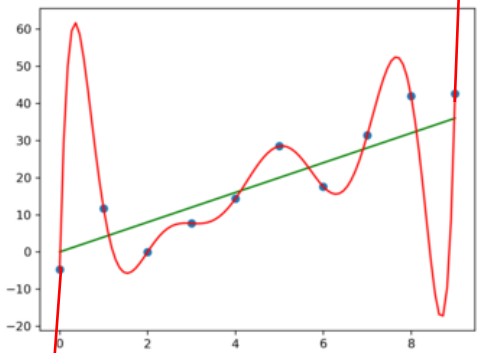
Overfitting only shows on unseen samples



The decision tree memorised your individual zodiacs



We did not learn *"penguinness"* but just our particular penguins



We cannot predict anything new

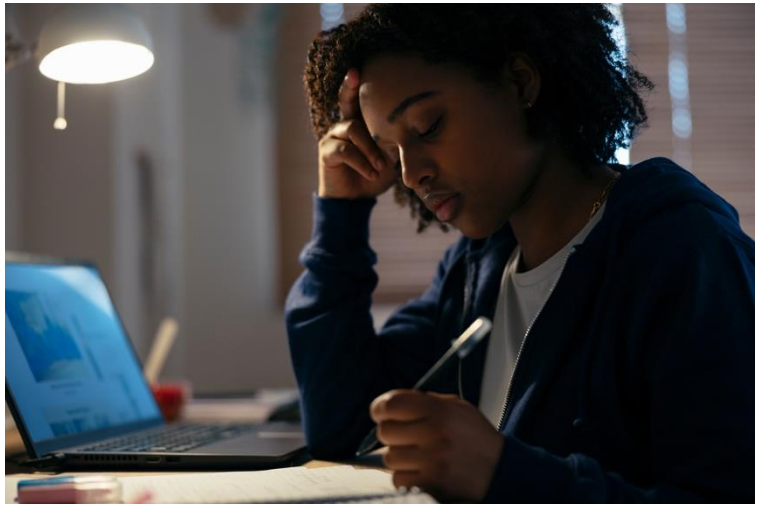
Overfitting just looks so darn good



You can overfit your cross-validation

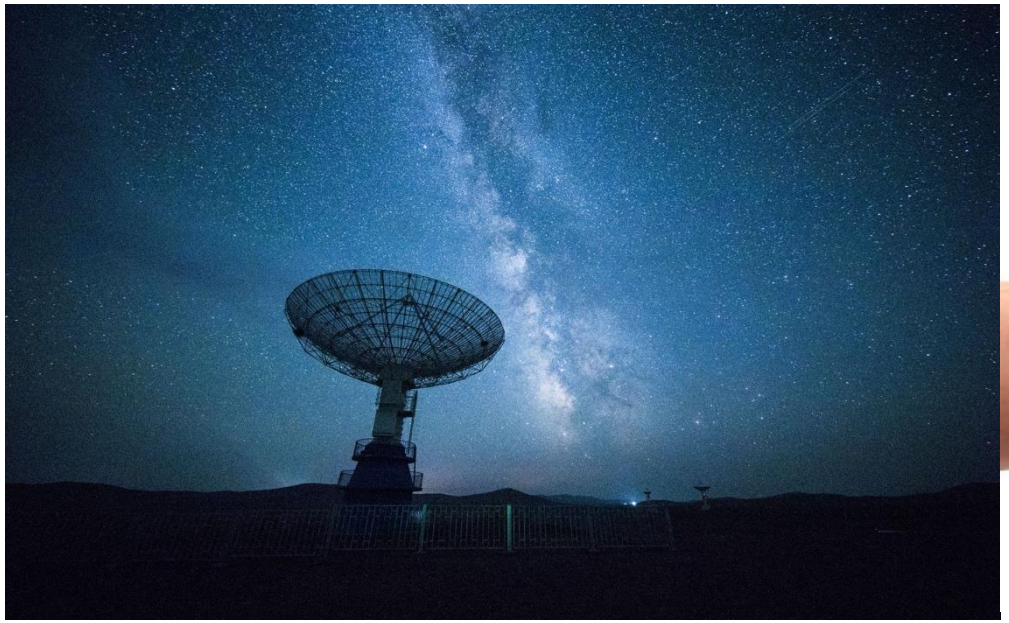


You can overfit by complaining to your colleague



You can overfit by telling your master's student they have to hand in soon

Mathematically proven ways to avoid overfitting optimism



Infinitely big sample size



A model that does not learn

We have some tools to fight back against overfitting



Regularization to constrain the models



Held out validation data such as cross-validation and test sets

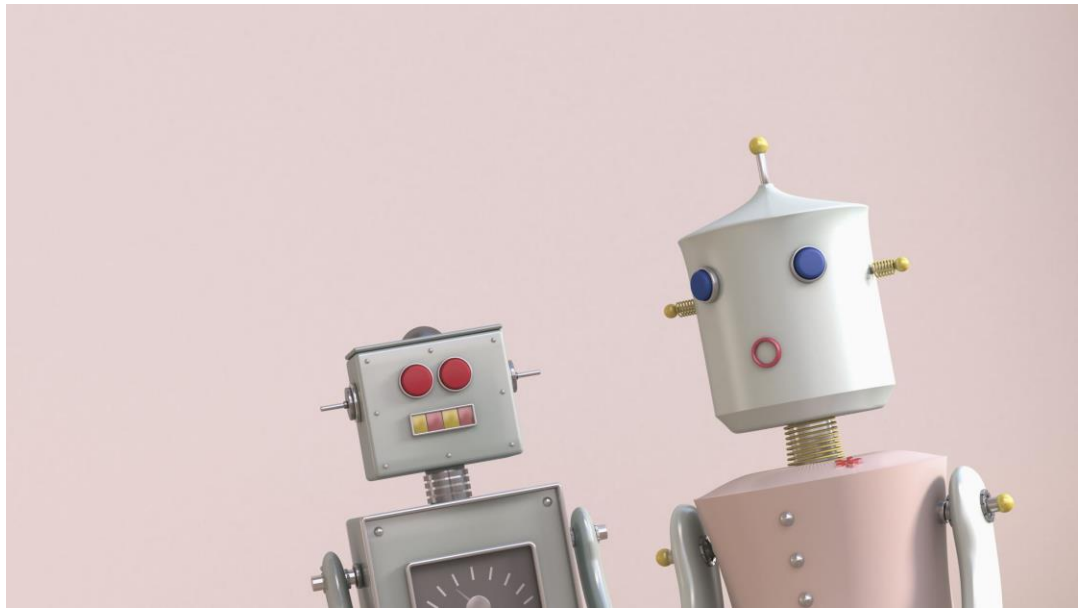


Bias-variance trade-off lets you trade model flexibility and bias. You can’t win.

Unfortunate truths about overfitting



Unfortunately, molecules are extra easy to overfit



“Hey Claude, are you sure we are not overfitting right now?”

Finally you got around to the forests you have literally been teasing for the past hour, dude



Remember how the decision tree saw all features and samples?



It was particularly keen on overfitting



Hear me out, let's just add more trees

Three rules for building forests



For one tree, we only look at a random subset of the samples

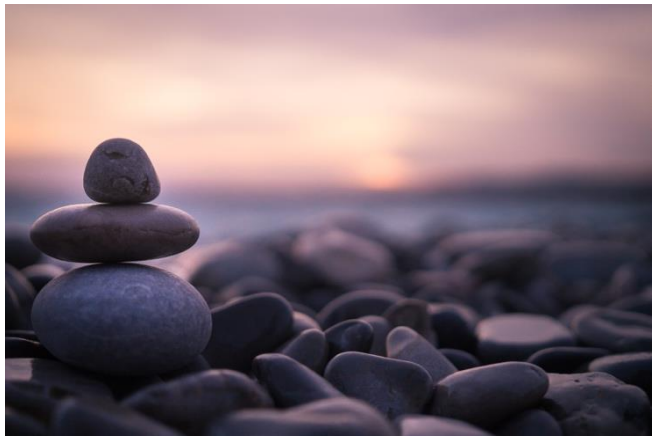


For one tree, we only look at a random subset of the features



In the end, each tree votes for the final label

Congratulations, we just invented Random Forests



Taking the average stabilises the variance

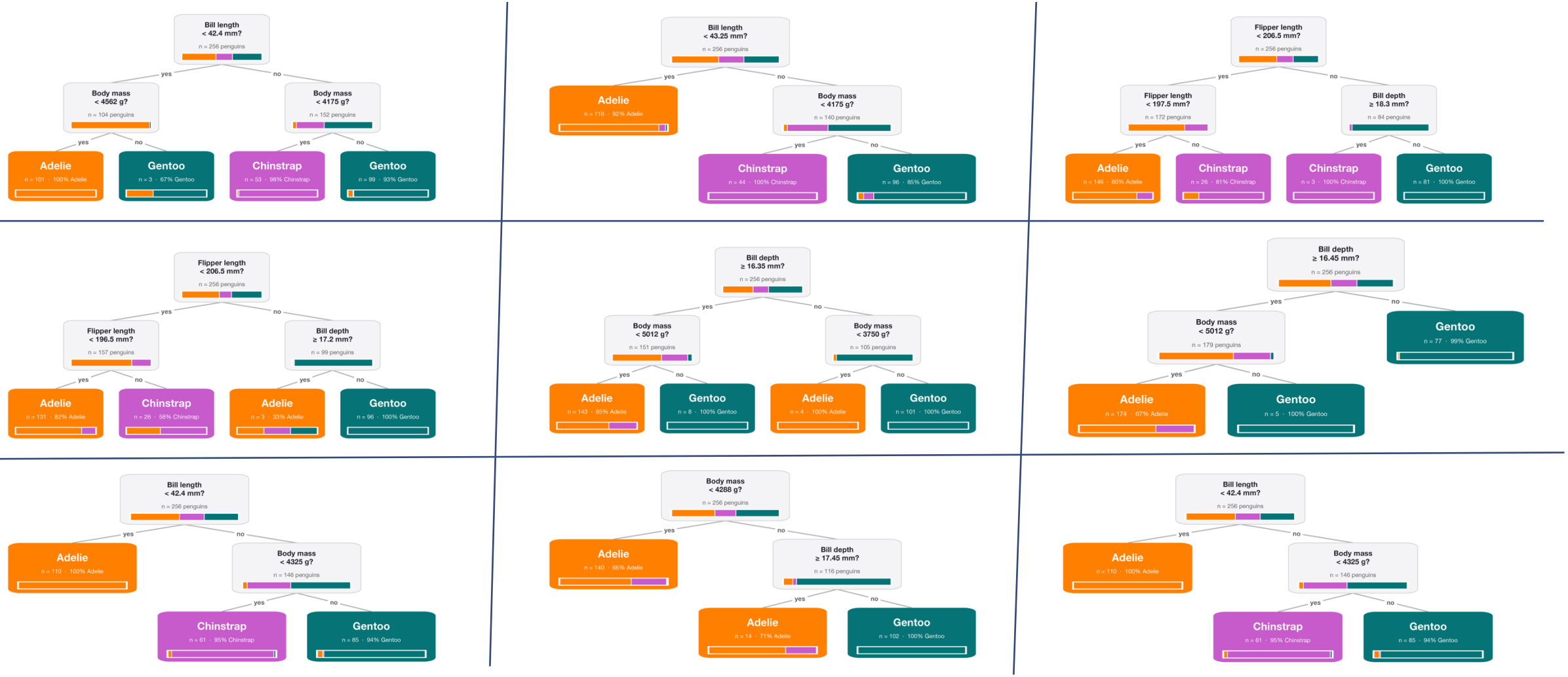


For one tree, we only look at a random subset of the features



You get feature importance for free

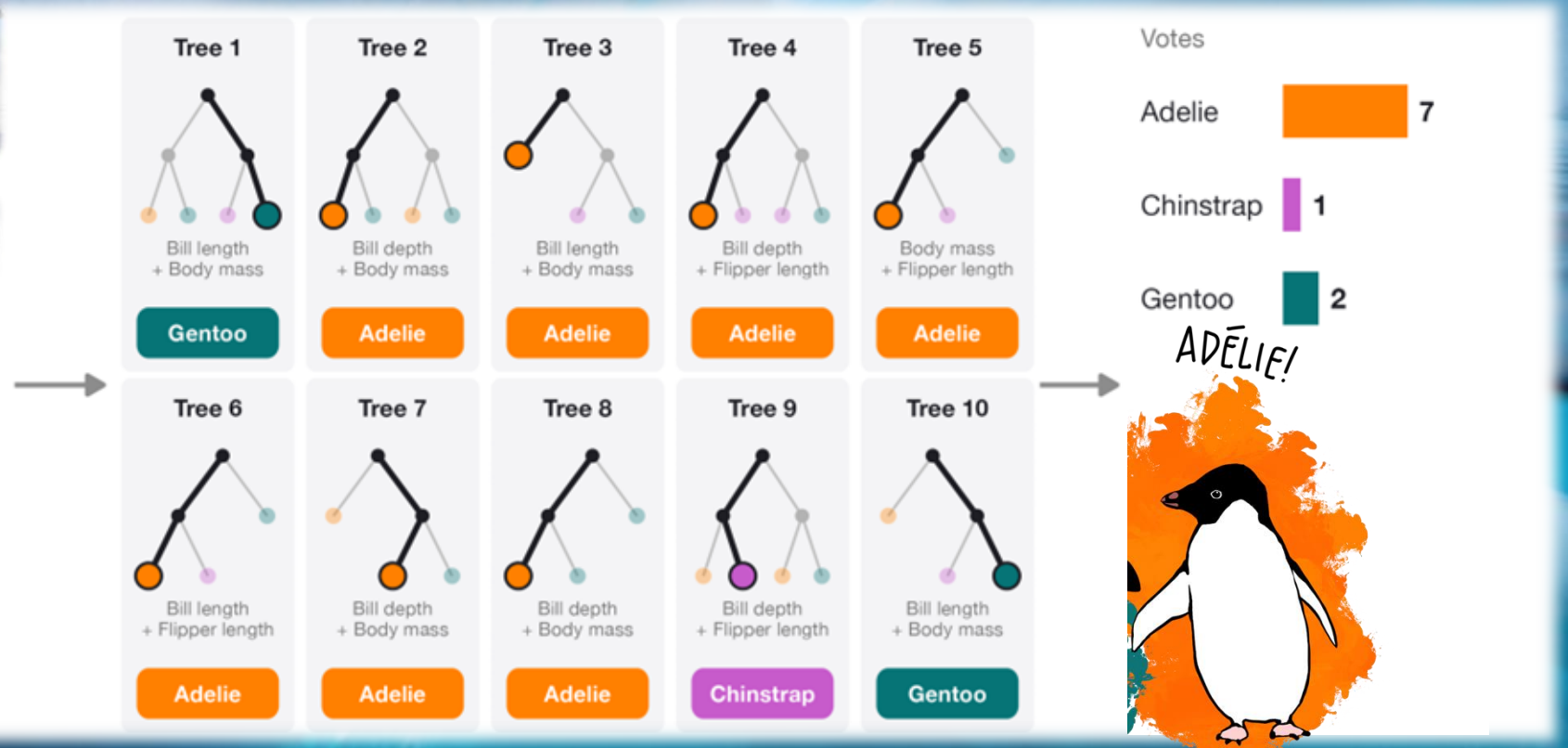
Building more forests for penguins



A penguin in a forest



Bill length	42.5 mm
Bill depth	20.7 mm
Flipper length	197 mm
Body mass	4500 g
Species	?



Afternoon Workshop



It is going to be cool



It is going to be something with molecules



and Random forests

Brief summary 2.0



Trees are neat and a good place to start



You can run but you cannot hide from overfitting



The robots forget overfitting exists



Random forests are very useful



The Afternoon Workshop is so fun

What are did we going to "learned"?

- ~~What is Predicting?~~
- ~~What does Learning even mean?~~
- ~~What are those, Trees anyway?~~
- ~~Why is Overfitting so very very scary?~~
- ~~Are Forests really that cool?~~
- What are we going to learn in this afternoon's Workshop?
- (Boosting trees)



That was all



Training samples have features and labels



Models predict labels from features



Training minimizes loss



Prediction puts labels on unseen samples



People still very much do this thing



Trees are neat and a good place to start



You can run but you cannot hide from overfitting



The robots forget overfitting exists



Random forests are very useful



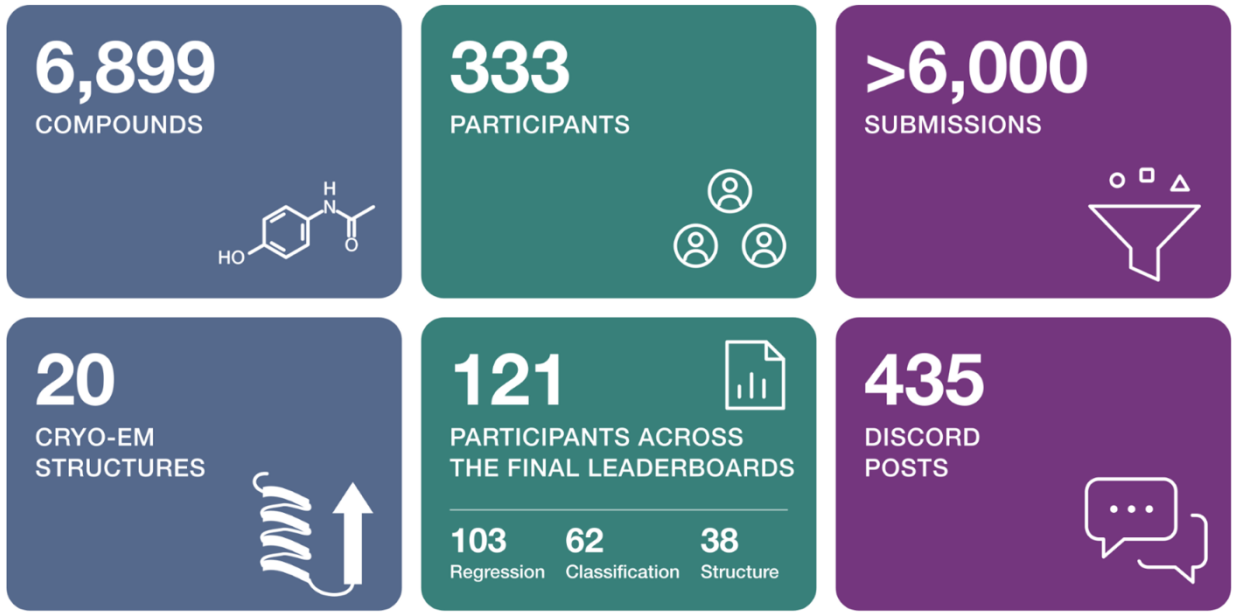
Boosted trees go brrrrr

Who won the competition?

Take Me Heme, Binding Modes: Wrapping up the CYP Inhibition Challenge

Let's have those country roads take us to the finish line! After 2 months and more than 6,000 submissions, we finally have our winners.

05 Nov 2026 Share



https://openadmet.ghost.io/take-me-heme-binding-modes-wrapping-up-the-cyp-inhibition-challenge/

Who won the competition?



Prediction of CYP450 Time dependent inhibition



Gradient-boosted trees dominated



Three top-performing algorithms* are all boosting trees

https://huggingface.co/spaces/openadmet/cyp-challenge

Gradient-boosted trees



Random forest fits many good trees in parallel

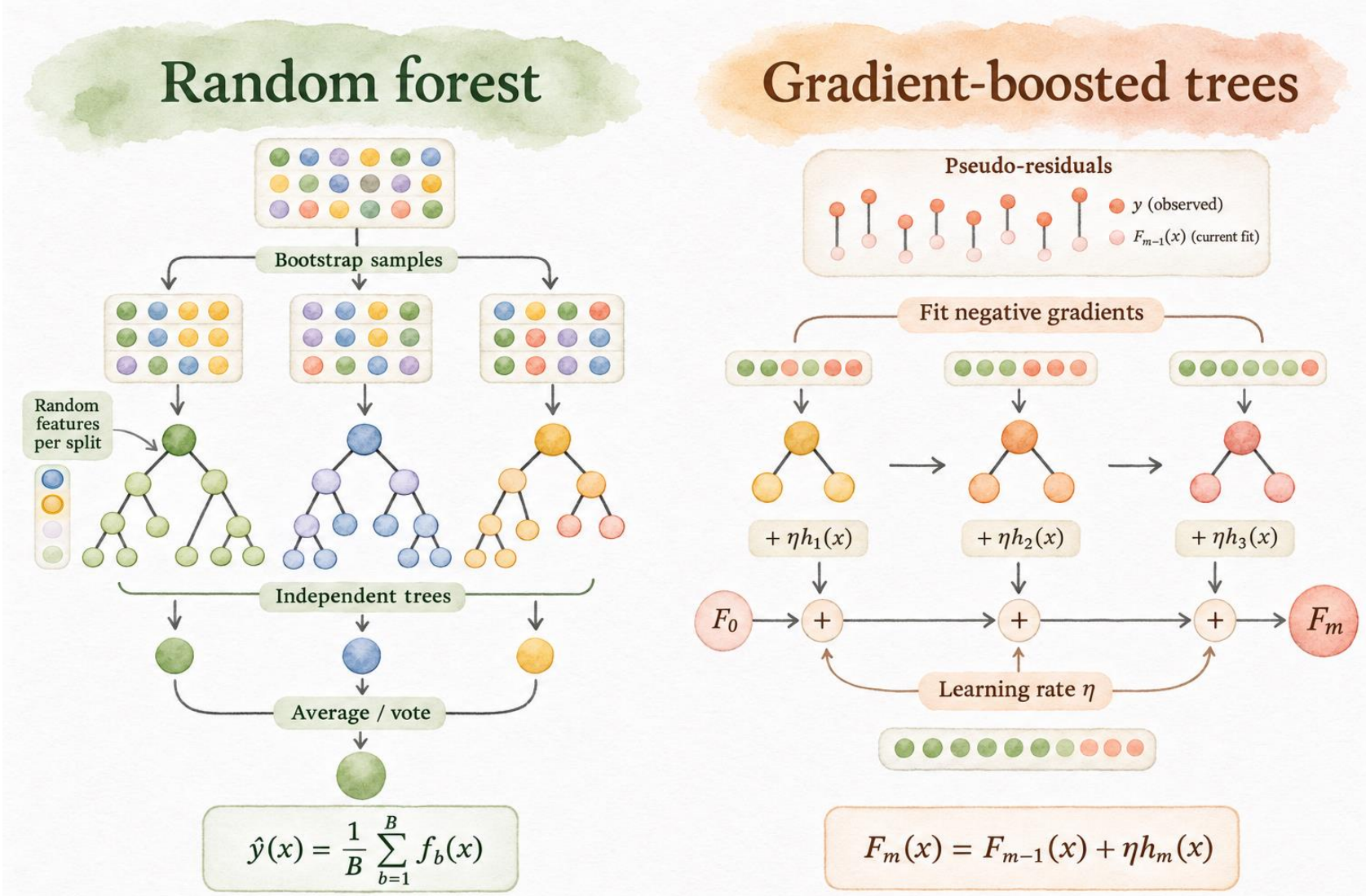


Boosted trees fit many weak-learners sequentially

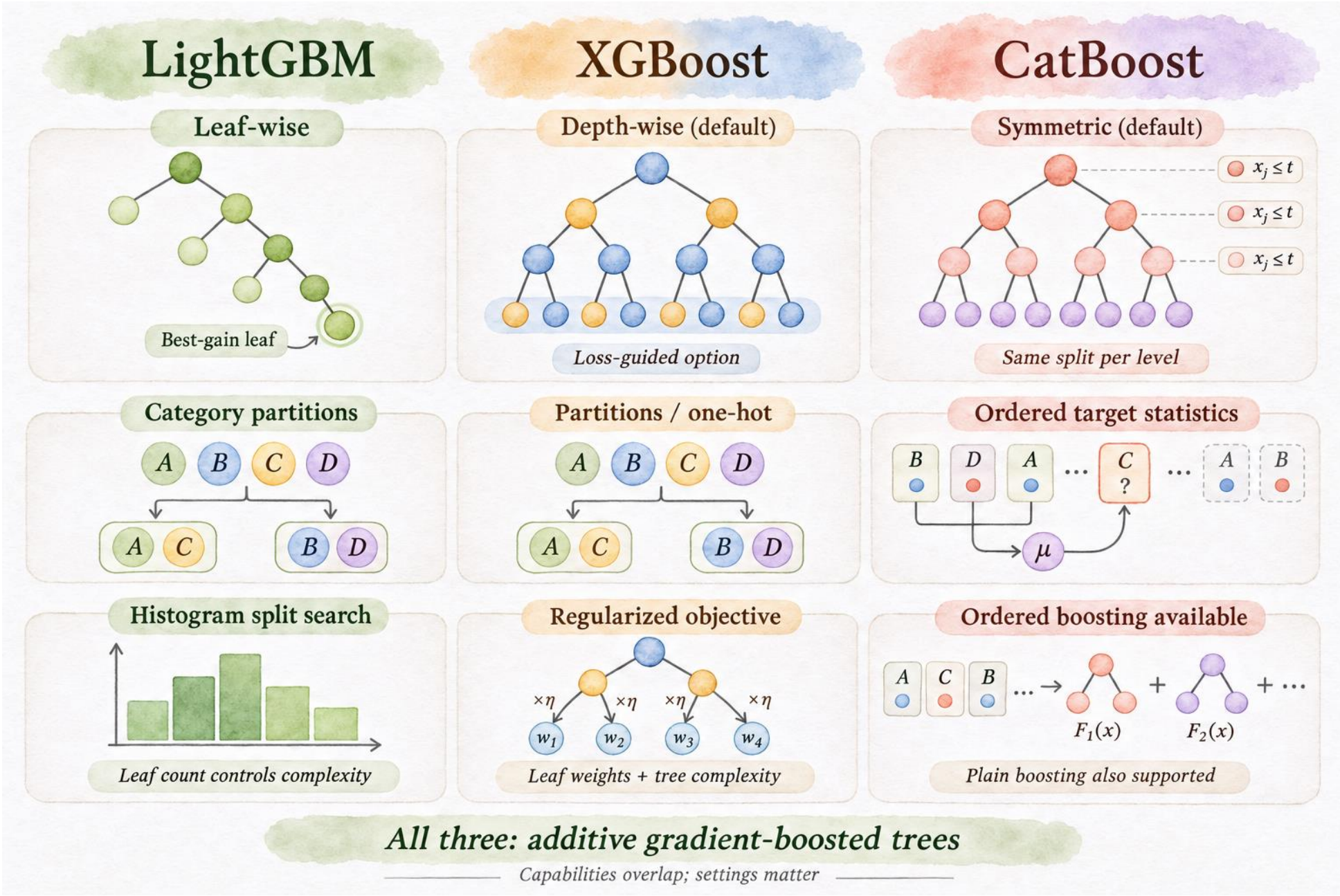


Each tree learns the error of the tree before it

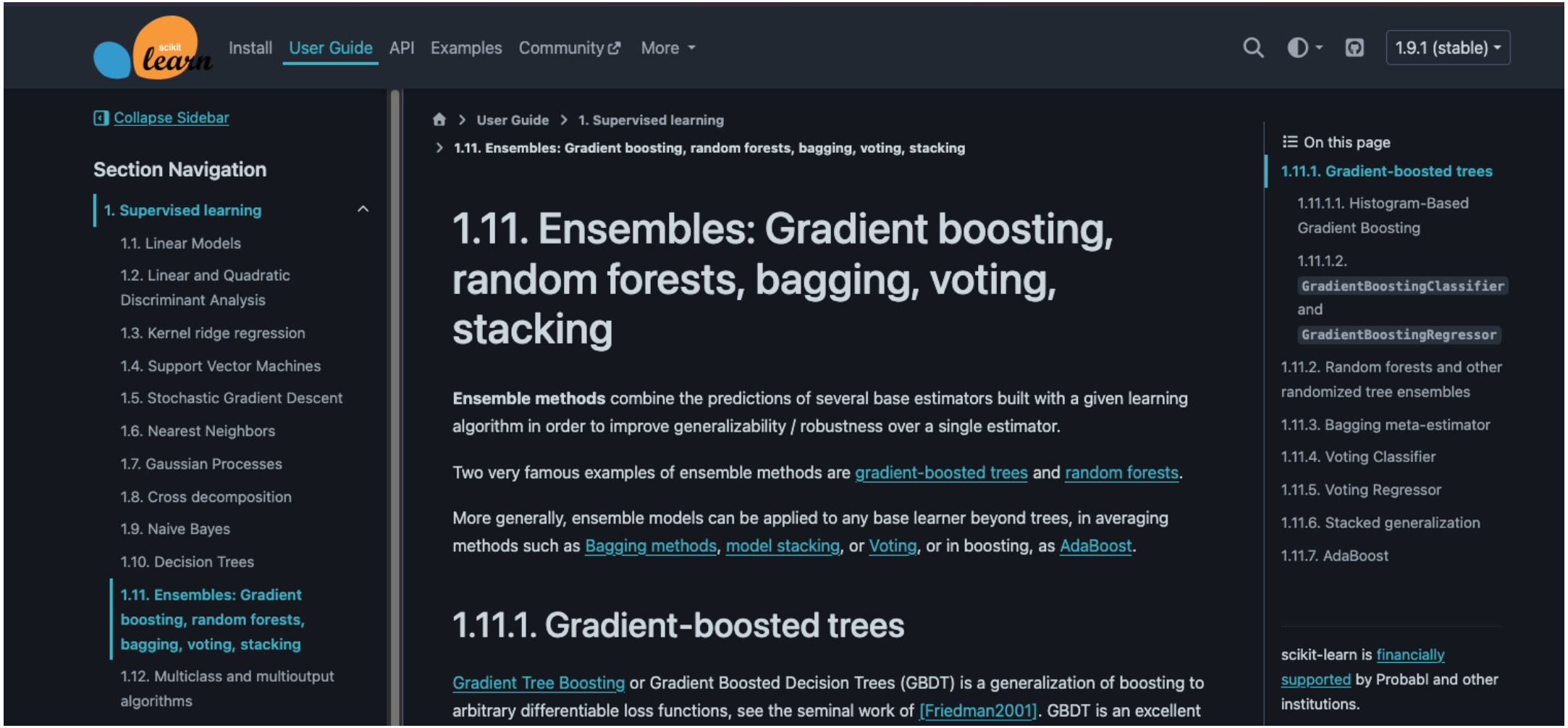
[BONUS] Gradient-boosted trees AI SLOP



[BONUS] Popular Boosting Algorithms AI SLOP



You can be king of the forest today! NOT AI SLOP



https://scikit-learn.org/stable/modules/ensemble.html